

User Manual

Plant Disease Prediction System

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1. Getting Started

About the System

- AI-based web application for plant disease prediction
- Uses symptom analysis to identify diseases
- No registration or login required for general users
- Accessible on any device with a browser and internet connection

System Requirements

Requirement	Specification
Device	Mobile, Tablet, or Computer
Browser	Google Chrome, Firefox, or Edge
Internet	Basic 2G/3G or Wi-Fi
Camera	Optional — for image upload only

How to Access

1. Open any web browser on your device
2. Type the application URL in the address bar
3. Press Enter — the home page loads instantly
4. System is ready to use — no setup required

2. User Roles

The system supports two user roles:

2.1 Farmer / General User

- No login required
- Access system directly from browser
- Enter plant symptoms and receive instant prediction
- Upload crop image (optional)
- View predicted disease, confidence score, and treatment suggestions
- Submit new prediction anytime using Predict Again

2.2 Admin

- Login required with authorized credentials

- Add, edit, or delete disease records
- Manage symptom checklist
- View all prediction logs
- Monitor system usage and performance statistics

Feature	Farmer / General User	Admin
Login Required	No	Yes
Submit Prediction	✓	✓
Upload Crop Image	✓	✓
View Treatment Suggestions	✓	✓
Manage Disease Records	✗	✓
View Prediction Logs	✗	✓
Manage Symptom Checklist	✗	✓
Monitor System Statistics	✗	✓

3. For Farmers / General Users

3.1 Accessing the System

- Open any web browser on your device
- Enter the application URL in the address bar
- Home page loads with the symptom input form
- No login required — start using immediately

3.2 Entering Plant Symptoms

- Locate "**Enter Plant Symptoms**" section on the home page
- Observe your plant carefully before selecting symptoms
- Select all symptoms that are clearly visible from the checklist:
 1. **Leaf Appearance** — Yellow Leaves, Brown Spots, Dark Spots, Pale Green Color
 2. **Surface Condition** — Powdery Coating, Wet Lesions, Dry Patches, Fungal Growth
 3. **Structural Changes** — Wilting, Curling, Leaf Drop, Stunted Growth
 4. **Other Indicators** — Foul Smell, Root Decay
- Select only what is clearly visible — do not guess
- Click "**Predict Disease**" button when done

3.3 Uploading a Crop Image (Optional)

1. Locate "**Upload Crop Image (Optional)**" below the symptom checklist
2. Click the upload area or drag and drop your image
3. Supported formats: JPG, JPEG, PNG
4. Maximum file size: 5 MB
5. Ensure image clearly shows the affected plant part

6. Skip this step if no clear image is available — symptom input alone is sufficient

3.4 Understanding Your Results

After submitting, the following information will be displayed:

- **Detected Disease** — Name of the most probable disease
- **Confidence Score** — System's certainty level about the prediction
 - a) Above 85% — Reliable, act on treatment recommendation
 - b) Below 70% — Consult a local agricultural expert
- **Disease Description** — Brief overview of the identified disease
- **Preventive Measures** — Steps to prevent further spread
- **Recommended Treatment** — Suggested fungicides, pesticides, or organic remedies

3.5 Submitting Another Prediction

1. Click "**Predict Again**" on the results page
2. Symptom form resets automatically
3. Enter new symptoms and submit

4. For Administrators

4.1 Accessing the Admin Panel

1. Open the application in any web browser
2. Add **/admin** to the application URL
3. Enter authorized **Username** and **Password**
4. Click "**Login**"
5. Admin Dashboard will open

Note: Three consecutive failed login attempts will temporarily lock the account. Contact the system developer to restore access.

4.2 Admin Dashboard Overview

Section	Purpose
Disease Management	Add, edit, or delete disease records
Symptom Management	Update symptom checklist
Prediction Logs	View all past predictions
System Statistics	Monitor usage and performance

4.3 Managing Disease Records

Adding a New Disease

1. Click "**Disease Management**" from the Dashboard
2. Click "**Add New Disease**"

3. Fill in all required fields:
 - Disease Name
 - Affected Crops
 - Associated Symptoms
 - Severity Level — Low, Moderate, or High
 - Disease Description
 - Preventive Measures
 - Treatment Options
4. Click **"Save Disease"**

Editing a Disease Record

1. Locate the disease in Disease Management
2. Click **"Edit"**
3. Update the required fields
4. Click **"Update"** to save changes

Deleting a Disease Record

1. Locate the disease record
2. Click **"Delete"**
3. Confirm deletion when prompted

Note: Deleted records cannot be recovered. Proceed with caution.

4.4 Managing the Symptom Checklist

Adding a New Symptom

1. Click **"Symptom Management"** from the Dashboard
2. Click **"Add New Symptom"**
3. Enter symptom name
4. Assign to group — Leaf Appearance, Surface Condition, Structural Changes, or Other Indicators
5. Click **"Save"**

Editing or Removing a Symptom

1. Locate the symptom in the list
2. Click **"Edit"** to rename or reassign — then click **"Update"**
3. Click **"Delete"** to remove — confirm the action

Note: Do not delete a symptom linked to any disease record — it may affect prediction accuracy.

4.5 Viewing Prediction Logs

1. Click **"Prediction Logs"** from the Dashboard
2. View the following for each prediction:

Column	Description
Log ID	Unique prediction identifier
Symptoms Entered	Symptoms selected by the user
Predicted Disease	Disease identified by the system
Confidence Score	Prediction certainty percentage
Date and Time	Timestamp of the prediction
Image Uploaded	Yes or No

3. Use search and filter to sort by date, disease, or confidence score
4. Review predictions with consistently low confidence scores

4.6 Monitoring System Statistics

1. Click "**System Statistics**" from the Dashboard
2. Review the following:

Statistic	Description
Total Predictions	Cumulative count of all predictions
Most Predicted Disease	Most frequently identified disease
Average Confidence Score	Mean confidence across all predictions
Average Response Time	System response time in seconds
Daily Active Users	Number of unique users per day

4.7 Logging Out

1. Click your username at the top right of the Dashboard
2. Select "**Logout**"
3. Login page will appear
4. Close the browser tab for additional security

5. Tips and Best Practices

5.1 For Farmers / General Users

- Observe your plant carefully before selecting symptoms
- Select only symptoms that are clearly and visibly present — do not guess
- Check multiple plants before entering symptoms for a more reliable result
- The more accurately you describe symptoms, the more reliable the prediction
- Upload a clear image in natural daylight for better prediction confidence
- Act immediately on predictions with confidence score above 85%
- Consult a local agricultural expert if confidence score is below 70%
- Use the system regularly — early detection reduces crop loss significantly

5.2 For Administrators

- Review and update the disease database at least once every season

- Verify symptom information from at least two reliable sources before adding a new disease
- Monitor prediction logs weekly to identify disease outbreak patterns
- Run a test prediction after every database update to confirm changes are working correctly
- Never share admin login credentials with unauthorized individuals
- Always log out completely after each admin session
- Contact the development team immediately if any system error or unusual behavior is noticed

6. Troubleshooting

6.1 For Farmers / General Users

Problem	Solution
Page not loading	Check internet connection and refresh browser
Symptom checklist not visible	Refresh page or update browser to latest version
Predict button not responding	Ensure at least one symptom is selected, then retry
Image upload not working	Check file format (JPG, JPEG, PNG) and size (max 5 MB)
Unexpected prediction result	Re-enter symptoms carefully — select only clearly visible ones
Low confidence score	Consult a local agricultural expert for professional diagnosis
Page displaying error message	Wait 30 seconds and resubmit — if issue persists, refresh browser

6.2 For Administrators

Problem	Solution
Unable to login	Check credentials carefully — passwords are case-sensitive
Account locked	Contact system developer to restore access
Disease not appearing in predictions	Verify at least one symptom is linked to the disease record
Prediction logs not updating	Refresh the logs page and check date filter settings
Changes not reflecting in application	Clear browser cache, log out, log back in and retest
System response time increased	Check System Statistics — contact developer if issue persists

6.3 Common Error Messages

Error Message	Cause	Solution
"Unable to connect to server"	Internet connection issue	Check connectivity and refresh
"No symptoms selected"	Form submitted without input	Select at least one symptom
"File format not supported"	Wrong image file type	Use JPG, JPEG, or PNG only
"File size too large"	Image exceeds 5 MB	Compress image and re-upload
"Prediction failed. Please try again."	Temporary server issue	Wait 30 seconds and resubmit

7. Need Help / Contact

7.1 Development Team

Name	Role	Email
Mo. Furqan Qureshi	Developer	furqanqureshi231005@acropolis.in
Sudarshan Rajput	Developer	sudarshanrajput230850@acropolis.in
Yogesh Mali	Developer	yogeshmali230064@acropolis.in
Zoya Qureshi	Developer	zoyaqureshi231171@acropolis.in

7.2 Faculty Supervisors

Name	Designation	Email
Prof. Mahendra K. Verma	Project Guide	mahendraverma@acropolis.in
Prof. Pawan K. Gupta	Project In-charge	pawangupta@acropolis.in

7.3 GitHub Repository

- **Link:** <https://github.com/sudarshanr04/EARLY-DISEASE-DETECTION.git>
- To report a bug or suggest improvement:
 1. Visit the repository link
 2. Click **"Issues"** tab
 3. Click **"New Issue"**
 4. Describe the issue clearly
 5. Click **"Submit New Issue"**

7.4 Institution

Detail	Information
Institution	Acropolis Institute of Technology and Research
Department	Information Technology
Location	Indore, Madhya Pradesh, India
Website	www.acropolis.in

